

Fans to Friends: Human Connectedness and Social Belonging by Transnational Live streaming Platform Design

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Abstract

Casual live streaming sessions (Casual Lives), often unedited and unscheduled, enhance the perceived “liveness” of streamers, promoting direct interactions between streamers and viewers, as well as vicarious connections among viewers. However, most platforms remain streamer-centric, offering limited support for direct viewer-to-viewer engagements. To address this gap, this study adopts a four-phase design process, prototype development, iterative refinement, user testing, and knowledge dissemination, presenting a high-fidelity prototype that synchronizes live content with multiple chat spaces calibrated to varying levels of perceived social distance. Drawing on the Casual Lives of South Korean popular music (K-pop), the study examines how purposeful platform design mediates fans’ experiences of homophily and belonging driven by parasocial experiences. This research seeks to advance the understanding of real-time social interaction by offering design implications that cultivate more inclusive, meaningful, and sustained forms of fan-to-fan connectedness within transnational communities.

CCS Concepts

• **Human-centered computing** → **User interface design; Activity centered design; Interface design prototyping.**

Keywords

Virtual Community, Fandom, Parasocial Relationships, Vicarious Parasocial Interactions, Live Streaming, Real-time Connectedness

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1 Introduction

Live streaming platforms (i.e., Instagram Live, YouTube Live, Twitch) introduce real-time communication affordances, shifting audience

participation from traditional one-way, nonreciprocal communication to a “one-and-a-half-way” interaction that enhances viewers’ perceived connectedness with media figures [17]. Casual live streaming sessions (Casual Lives), typically unscheduled and streamed in relaxed settings, such as private rooms or backstage areas, further deepen viewers’ sense of connectedness while fostering their intense, imagined and non-reciprocal affection for media figures [6, 18, 21, 25–27]. In the context of South Korean popular music (K-pop), frequent and carefully curated Casual Lives amplify the “on-screen authenticity” of K-pop musicians (idols) by offering fans intimate access to personal stories and candid moments of their everyday lives [16]. This constructed authenticity becomes a powerful means for idols to build affinity with fans worldwide, forming parasocial bonds that underpin the emotional investment and engagement of the global K-pop fandom [15, 16, 22].

Beyond fostering parasocial bonds, K-pop Casual Lives also facilitate vicarious parasocial interactions among fans, mediated through the use of integrated chat features. The text-based digital converses transform Casual Lives from individual parasocial indulgence into a collective social experience, strengthening the shared sense of community and belonging among fans [12]. These real-time social engagements offer valuable emotional capital, including social acceptance, companionship, and peer recognition, which helps mitigate feelings of boredom, loneliness, and isolation [29].

Using K-pop Casual Lives as the research context, this study hypothesizes that addressing fans’ social needs through thoughtful platform design can foster meaningful real-time fan-to-fan interactions. Particularly, it investigates platform affordances that embody varying degrees of social distances, with the aim of activating a more intensive homophily effect (“birds of a feather”) within smaller fan groups [19]. By leveraging this effect, such affordances are anticipated to prompt diverse patterns of fan-to-fan interaction while reinforcing the real-time sense of connectedness and belonging. The study poses the following research question: “How can platform affordances reflecting varying degrees of social distance foster meaningful interactions between fans during transnational K-pop Casual Lives?”

This research adopts a four-phase approach: (1) the development of a high-fidelity pilot prototype featuring multiple live streaming chat spaces (see Section 4 for further details on the concept and development); (2) iterative refinements of the prototype, informed by feedback from focus groups and the HCI community during demonstration sessions at academic conferences; (3) usability testing to evaluate usage behavior and chat content facilitated by the refined prototype; (4) paper preparation and knowledge dissemination. By

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Figure 1: Prototype Interface (work in progress). Right: Youtube Live chat and Private Chat consolidated in the multi-chat area. Left: live streaming content.

offering empirical insights into real-time fan interactions, this research aims to advance the CSCW community’s understanding of the real-time “platform society” [28] — a digitally mediated collective formed through shared affinities, dynamic interactions, and fan identity. In addition, it provides insights into design implications that foreground the creation of more meaningful forms of digital companionship.

2 Background

Over the past two years, our research has examined technology-mediated real-time interactions within global K-pop fandoms. We found that many fans, even those with little to no proficiency in Korean, participate in Casual Lives not only to engage with their favorite idols but also to interact with fellow fans. Prior work indicates that *Social Fans* seek to build connections with others who share their K-pop enthusiasm, valuing the sense of community and belonging that fandoms provide. These fans often form more meaningful connections with others from the same fandom, a phenomenon likely explained by homophily — the tendency to appreciate like-minded individuals [7, 19]. This finding resonates with Ham and Lee [9], who emphasize the central role of fan identity in fostering interpersonal connections within fandoms: the stronger a fan identifies with their fan identity, the greater the satisfaction they perceive from fan-to-fan interactions.

To navigate the complexities of the multilingual and multicultural environment presented by transnational K-pop Casual Lives, coupled with the fast-paced and large-scale influx of live chat messages, *Social Fans* often rely on multiple screens or devices to interact with one another in real time. Some fans create subgroup chats on external social media platforms or applications (e.g., Facebook Messenger, WhatsApp) to conduct private conversations with “fan friends,” fellow fans from the same K-pop fandom with whom they share more personal or intimate connections. Others prefer staying in the live chat, passively viewing or selectively engaging

with messages from more distant fellow fans. However, the reliance on multiple applications and devices fragments the attention of fans, resulting in significant cognitive stress and frustration that decrease the overall enjoyment of their live streaming experience.

3 Related Work

3.1 Platform Design

Online communities, including fandoms, have drawn significant attention from HCI scholars, who explore the role of technology in shaping digital communities, establishing social norms, and facilitating social collaboration [1, 3–5, 22, 23]. Ringland et al. [22], among others, contend that “platforms, and their affordances, shape and constrain how online communities engage in their practices” (p.3). Similarly, Van Dijck et al. [28] argue the pivotal role of platform design in structuring social interactions by embedding particular agendas, norms, and values into the architecture. For instance, VLive, a discontinued K-pop-centric live streaming platform designed for global audiences [8], supported community participation through affordances such as fan-subtitling and fan-generated photo reels [15].

3.2 Transnational Fandom

The global K-pop fandom comprises international fans residing outside South Korea, forming hybrid digital networks that transcend linguistic and cultural boundaries through shared interests, strong emotional investment, and collective social identity [20, 30]. These fandoms are distinctive for their ability to enable both intra- and inter-fandom collaborations, overcome barriers via fan-driven services, and engage in meaningful social activism on behalf of idols [13, 22]. Such collective endeavours extend beyond amplifying fandom visibility; they reinforce group identification and cultivate reciprocal trust, thereby fostering cohesive communities grounded in shared purposes and collective values [14].

3.3 Parasocial Experiences

Parasocial interaction (PSI) refers to a one-sided, imagined form of interpersonal exchange, constructed through the performative behaviours of media figures, such as facing the camera and sustaining eye contact [10]. PSI generates a potent ‘illusion of intimacy,’ prompting audiences to imagine reciprocity and interactivity, and to engage with media figures as if in mutual relationships. Parasocial relationships (PSR), extending from the notion of PSI, embody an intensified form of simulated intimacy that is akin to close relationships such as friendship, partnership, and kinship [6, 20, 24, 26, 30].

As social media became increasingly prevalent and ubiquitous, scholars began reexamining PSI and PSR through a digital social lens, exploring how these parasocial phenomena manifest and evolve in computer-mediated contexts.

Vicarious parasocial interaction (VPSI), introduced by Dai and Walther [2], was first identified on Twitter (now X), where spectators develop a sense of parasocial closeness or intimacy by observing text-based interactions between a media figure and their followers. Through such observations, spectators learn the “know-how” of engagement, acquiring digital etiquette, tone, and mannerisms that could facilitate their own interactions with the media figure. The perceived intimacy is further amplified when spectators recognize shared attributes with the observed followers, such as nationality or personal interests. Building off Dai and Walther, Martin and Cohen [18] argue that during live streaming sessions, spectators frequently alternate between direct PSI and VPSI, perceiving both as comparable forms of pseudo-interactions. Similarly, Speed et al. [23] identify vicarious enjoyment as a central motivator in Twitch’s Gaming Lives, where “loyal lurkers” regularly attend sessions without directly interacting with the streamer.

4 Prototype Concept and Design

To address the social needs of global fans during live streaming sessions, we developed a high-fidelity prototype that mitigates a critical gap in current live streaming platforms: their inadequate support for meaningful fan-to-fan interaction. This limitation not only compels fans to navigate multiple applications or devices to engage with smaller subgroups, but also leads some to withdraw from social participation altogether to avoid the cognitive stress stemming from multitasking. Therefore, this prototype repurposes a live streaming platform as a temporal social venue, capable of creating meaningful interactions within transnational fan communities.

To align the prototype design with established user practices, we analyzed the user interface and user experience of YouTube, a live streaming platform widely used by global K-pop fans. Based on this analysis, we integrated both the YouTube Live Chat and a Private Chat component into the prototype interface (see Figure 1), presenting multiple scales of social space and participation that allow users to shift from a broad transnational community to a more intimate, exclusive space. Moreover, Private Chat represents the small-group dynamics that Social Fans currently sustain across multiple external apps [11]. This integration facilitates fans’ seamless engagement in both vicarious parasocial and direct social interactions during K-pop Casual Lives.

By empowering fans to create their own private chat rooms and selectively invite their fan friends, the feature seeks to foster more

meaningful and enduring conversations. Personalization functions, such as the ability to change profile and chat room pictures, adopt aliases, and display personal statuses, further strengthen fans’ sense of security and community. The YouTube Live Chat resembles a large stadium concert, where fans are immersed in the excitement of a vast, diverse crowd. In contrast, Private Chat is akin to a cozy home gathering, where a close-knit group of friends share the same live streaming experience with greater intimacy, trust, and mutual recognition. By offering both large-scale communal enthusiasm and small-scale personal connection, the multi-chat affordance seeks to enhance fans’ sense of participation and enriches the quality of fan-to-fan interactions. This prototype embodies a purposeful platform design that fosters meaningful real-time social interactions among K-pop fans, thereby strengthening their connectedness and reinforcing a sustained sense of community belonging.

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